Practice Quiz: Body Science — Module 04: Cognition

Name: ___ ***Date**:* ___

Part A: Multiple Choice

1. Your brain is often described as a "prediction machine." This means:

A) It can see the future B) It constantly builds and updates mental models to anticipate what will happen next in your body and environment C) It only predicts when you're taking a test D) Predictions and thinking are unrelated

1. When your body "knows" to release digestive enzymes before food reaches your stomach (just from the smell of food), this is an example of:

A) A mistake B) Your body's cognitive system making a prediction based on past experience and acting ahead of time C) Digestion having nothing to do with the brain D) A random chemical reaction

1. Your brain uses about 20% of your body's energy. Most of that energy goes to:

A) Moving your muscles B) Running predictions and mental models that manage your body and make sense of the world C) Growing hair D) Storing fat

1. The placebo effect (feeling better after taking a sugar pill you think is medicine) shows that:

A) Medicine doesn't work B) Your brain's beliefs and predictions can change how your body actually feels and heals C) Sugar pills are real medicine D) The placebo effect isn't real

1. Your brain has to decide which body signals to pay attention to and which to ignore. This process is called:

A) Sleeping B) Attention — your brain's way of prioritizing the most relevant predictions C) Shutting down D) Random selection

1. When you learn that a dark alley is actually a safe, well-lit path, your fear decreases. This is because:

A) The alley physically changed B) Your brain updated its mental model of the situation, which changed its prediction from "danger" to "safe" C) Fear is not related to thinking D) Nothing in your brain changed

1. Phantom limb pain (where people feel pain in a limb that has been amputated) demonstrates that:

A) Pain only comes from the body B) Your brain maintains a model of your body, and it can generate sensations based on that model even when the body part is gone C) Amputated limbs grow back D) The brain has no model of the body

---

Part B: Short Answer

1. Explain the **placebo effect** using Active Inference ideas. How do your brain's predictions about a treatment actually change what happens in your body? Why does this show that cognition is not separate from your physical health?

2. Your brain makes predictions about your body that you're not aware of (like predicting your blood oxygen level). Describe two examples of **unconscious cognitive predictions** your body makes. What happens when those predictions are wrong?

3. Compare the way your brain handles a **familiar situation** (like walking to school) with an **unfamiliar situation** (like visiting a hospital for the first time). How does your brain's cognitive processing differ? Which situation uses more energy, and why?